Chengyu (Cathy) Fan

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Education

Colby College, Waterville, ME, USA May 2026
Bachelor of Arts
Double Major: Computer Science with AI Concentration (*Distinction in Major and Honors in Major*), Mathematical Sciences

Research Experience

Independent Research Jun 2026 – Present

- Exploring AI security and mechanistic interpretability

Research Assistant – Reliability in Multimodal Interaction Dec 2024 – May 2026
Human, MachiNe, and Environment (HUMANE) Lab, Colby College
Advisor: Prof. Tahiya Chowdhury

- Developing *PoseDoc*, an interactive tool that aims to help users inspect, refine, and customize human pose annotations more efficiently [P1]
- Classified common occlusion sources into four types and proposed the *Occlusion Index (OI)* for quantifying occlusion in human pose estimation [P2]

Independent Study – Accessible Art Tool Sep 2025 – May 2026
INSITE Lab, Colby College
Advisor: Prof. Stacy Doore

- Developing a multimodal database tool for accessible artwork descriptions for people with visual impairments.

Research Assistant – Conversation Analysis May 2024 – Dec 2024
Department of Computer Science and Psychology, Colby College
Advisors: Prof. Tahiya Chowdhury, Prof. Veronica Romero

- Quantified movement coordination in dyadic conversations [P3]

Research Assistant – LLM Agent Simulation Jan 2024 – Aug 2024
Davis Institute for Artificial Intelligence (Davis AI), Colby College
Advisor: Dr. Michael Yankoski

- Developed *Comp-HuSim*, a generative multi-agent simulation platform modeling complex human-like behaviors with 60+ AI agents [P4]

Teaching Experience

Teaching Assistant Feb 2023 – Dec 2025
Departments of Computer Science & Mathematics, Colby College

Provided academic support through grading and TA hours for a range of foundational and advanced courses, including: MA262 Vector Calculus, CS343 Neural Networks, CS333 Programming Languages, CS231 Data Structures & Algorithms, CS154 Natural Language Processing.

Work Experience

Founder & AI Security Researcher Jul 2026 – Present
Ptero AI Research LLC (*Self-employed*)
Early-stage exploration of AI security and mechanistic interpretability, including literature review and exploration of potential research directions.

Software Development Intern Dec 2023 – Jan 2024
STE Transmission (Subsidiary of Sany Heavy Industry Co., Ltd), Changshu, China
Developed an online platform to monitor power consumption of high-energy industrial machines.

Machine Learning Intern May 2023 – Aug 2023
Co-hosted by SureStart & Davis AI, Colby College
Supervised by Dr. Amanda Stent. Designed and evaluated a text-to-3D generative AI model.

Peer-Reviewed Publications

* indicates equal contribution.

[p1] **Fan, C.** & Chowdhury, T.(2025). **PoseDoc: An Interactive Tool for Efficient Keypoint Annotation in Human Pose Estimation.** In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25).*(Accepted, to appear.)

[p2] **Fan, C.** & Chowdhury, T.(2025). **When Pose Estimation Fails: Measuring Occlusion for Reliable Multimodal Interaction.** In *Companion Proceedings of the 27th International Conference on Multimodal Interaction (ICMI Companion '25).* (Accepted, to appear.)

[p3] **Fan, C.**, Romero, V., Paxton, A., & Chowdhury, T. (2024). **Towards Multimodality: Comparing Quantifications of Movement Coordination.** In *Companion Proceedings of the International Conference on Multimodal Interaction (ICMI Companion '24)*

[p4] **Fan, C.***, Tariq, Z.*, Bhuiyan, N. S., Yankoski, M. G., & Ford, T. W. (2024). **Comp-HuSim: Persistent Digital Personality Simulation Platform.** In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '24).*

Grants and Awards

- **Honorable Mention**, CRA Outstanding Undergraduate Researcher Award (Computing Research Association), 2025–2026
- Compagna-Sennett Iterate and Expand Grant ($4,000), Halloran Lab for Entrepreneurship, Colby College — awarded for the startup *InclusiM* with two co-founders (2025)
- Iteris Hackathon (Award-Winning Team, 2024); Davis AI / Dataiku Datathon (Creativity in Presentation Award, 2023), Colby College

Extracurricular Activities

Colby Robotics (Co-President) Fall 2023 – May 2026
Organize and lead workshops on soldering, 3D printing, CircuitPython, Git/GitHub, and Micromouse.